Jigsaw - Agile Community Rules Classification

赛后讲解

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赛题背景

竞赛任务介绍：文本二分类

该竞赛要求参赛者构建一个二分类模型，用于预测Reddit**评论是否违反特定规则**。数据集包含来自多个subreddit的已审核评论，并基于真实的subreddit规则。参赛者需使用一个标注的小开发集进行训练，测试集则包含一些未在训练集出现过的新规则。

- 数据字段：

字段	描述
body	评论的文本
rule	评论被判断违反的规则
subreddit	评论所在的论坛
positive_example_{1,2}	违反规则的评论示例
negative_example_{1,2}	不违反规则的评论示例
rule_violation	表示评论是否违反规则（1：违反，0：不违反）

赛题背景

数据概览

	row_id	body	rule	subreddit	positive_example_1	positive_example_2	negative_example_1	negative_example_2	rule_violation
0	0	Banks don't want you to know this! Click here ...	No Advertising: Spam, referral links, unsolici...	Futurology	If you could tell your younger self something ...	hunt for lady for jack off in neighbourhood ht...	Watch Golden Globe Awards 2017 Live Online in ...	DOUBLE CEE x BANDS EPPS - "BIRDS"\n\nDOWNLOAD/...	0
1	1	SD Stream [ENG Link 1] (http://www.sportsstre...	No Advertising: Spam, referral links, unsolici...	soccerstreams	[I wanna kiss you all over! Stunning!](http://...	LOLGA.COM is One of the First Professional Onl...	#Rapper \n🚨Straight Outta Cross Keys SC 🚨 YouTu...	[15 Amazing Hidden Features Of Google Search Y...	0
2	2	Lol. Try appealing the ban and say you won't d...	No legal advice: Do not offer or request legal...	pcmasterrace	Don't break up with him or call the cops. If ...	It'll be dismissed: https://en.wikipedia.org/w...	Where is there a site that still works where y...	Because this statement of his is true. It isn'...	1
3	3	she will come your home open her legs with an...	No Advertising: Spam, referral links, unsolici...	sex	Selling Tyrande codes for 3€ to paypal. PM. \n...	tight pussy watch for your cock get her at thi...	NSFW(obviously) http://spankbang.com/iy3u/vide...	Good News ::Download WhatsApp 2.16.230 APK for...	1
4	4	code free tyrande --->>> [Imgur](http://i.imgu...	No Advertising: Spam, referral links, unsolici...	hearthstone	wow!I amazing reminds me of the old days.Well...	seek for lady for sex in around http://p77.pl/...	must be watch movie https://sites.google.com/s...	We're streaming Pokemon Veitnamese Crystal RIG...	1
...	...	...	...	...	...	...	...	...	...
2024	2024	Please edit your post so it is readable. These...	No legal advice: Do not offer or request legal...	relationships	I'm not ok with this in anyway, and think you ...	See a lawyer under the guise that you want thi...	This is just untrue. OP is 13, not an adult -...	Why should I care about all the bicyclists I i...	1
2025	2025	Yes, and in a right to work state they can eve...	No legal advice: Do not offer or request legal...	legaladvice	Move as much of your assets as you can offshor...	We have great consumer protection laws. There'...	LPT piratebay, transmission, vpn. Get the musi...	It's not so much that I killed them, it's that...	0
2026	2026	**HD** Streams: |ENG **HD**[Watch here..PC & ...	No Advertising: Spam, referral links, unsolici...	soccerstreams	stitopdisca1987.tumblr.com - sex Take girl for...	this girl get sex going to to old http://mrk....	NO ADS | NO ADS | NO ADS\n\nWe show all SOCCER...	[So great! Thanks for sharing.](http://www.che...	1
2027	2027	No. Not when doing so obviously presents a saf...	No legal advice: Do not offer or request legal...	politics	SHE ISNT A BIRTHING CHAMBER BUT EQUALLY THE BA...	Jail? What are you, ten years old? If they pro...	Who cares about that when I can keep raping in...	send me a private message; I may be able to he...	1
2028	2028	> CA is an at fault state so they will not be ...	No legal advice: Do not offer or request legal...	legaladvice	[Yes, it is. In the USA, at least](https://www...	Never pay a civil demand. \n\nIt's a bluff and...	Where is there a site that still works where y...	Steal a car from an elderly person who won't n...	1

2029 rows × 9 columns

```
np.unique(train['subreddit'])
```

```
array(['2007scape', 'Android', 'AskHistorians', 'AskReddit',
       'AskTrumpSupporters', 'AskWomen', 'BlackPeopleTwitter', 'CFB',
       'CanadaPolitics', 'Christianity', 'DIY', 'DestinyTheGame',
       'EnoughTrumpSpam', 'Futurology', 'Games', 'GetMotivated',
       'GlobalOffensive', 'GlobalOffensiveTrade', 'IAmA', 'Incels',
       'LateStageCapitalism', 'LifeProTips', 'MMA', 'NSFW_GIF',
       'NeutralPolitics', 'OldSchoolCool', 'OutOfTheLoop', 'Overwatch',
       'PoliticalDiscussion', 'PurplePillDebate', 'SandersForPresident',
       'ShitRedditSays', 'Showerthoughts', 'SubredditDrama',
       'SuicideWatch', 'TheSilphRoad', 'The_Donald', 'TwoXChromosomes',
       'UpliftingNews', 'anime', 'askscience', 'asoiaf', 'atheism', 'aww',
       'books', 'canada', 'changemyview', 'churning', 'conspiracy',
       'creepyPMs', 'dataisbeautiful', 'depression', 'europe',
       'explainlikeimfive', 'fantasyfootball', 'food', 'funny',
       'gameofthrones', 'gaming', 'gifs', 'gonewild', 'hearthstone',
       'hillaryclinton', 'hiphopheads', 'history', 'india', 'jailbreak',
       'leagueoflegends', 'legaladvice', 'me_irl', 'movies', 'nba',
       'news', 'nfl', 'nosleep', 'nottheonion', 'pcmasterrace',
       'personalfinance', 'philosophy', 'photoshopbattles', 'pics',
       'pokemon', 'pokemongo', 'pokemontrades', 'politics',
       'relationships', 'science', 'sex', 'soccerstreams', 'socialism',
       'space', 'spacex', 'syriancivilwar', 'technology', 'television',
       'tifu', 'videos', 'whatisthisthing', 'worldnews', 'wow'],
      dtype=object)
```

Rule in training set：

- No Advertising: Spam, referral links, unsolicited advertising, and promotional content are not allowed.
- No legal advice: Do not offer or request legal advice.

赛题背景

评价指标：AUC

Evaluation

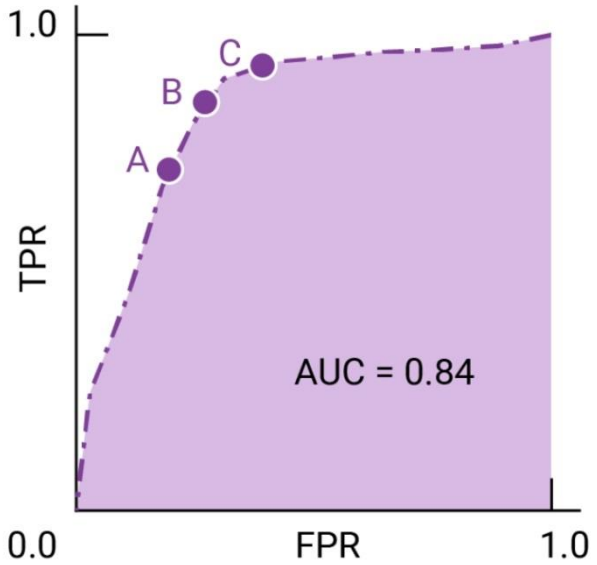
Scoring

Submissions are evaluated on column-averaged AUC.

Submission File

For each `row_id` in the test set, you must predict the probability that a comment violates a given rule. The file should contain a header and have the following format:

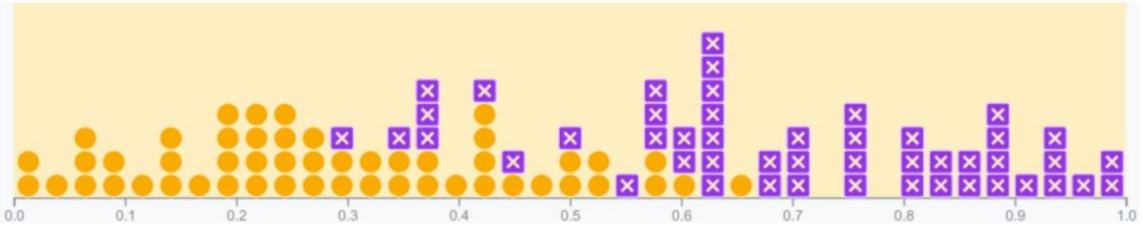
```
row_id,rule_violation
2029,0.5
2030,0.67
2031,0.1
etc.
```



曲线下面积 (AUC)

ROC 曲线下面积 (AUC) 表示，如果给定随机选择的正例和负例，模型将正例排在负例之上的概率。

上面的完美模型包含边长为 1 的正方形，其曲线下面积 (AUC) 为 1.0。这意味着，模型将随机选择的正例正确排在随机选择的负例之上的概率为 100%。换句话说，查看下方的数据点分布情况，AUC 可提供模型将随机选择的方形放置在随机选择的圆形右侧的概率，而不会受到阈值设置位置的影响。

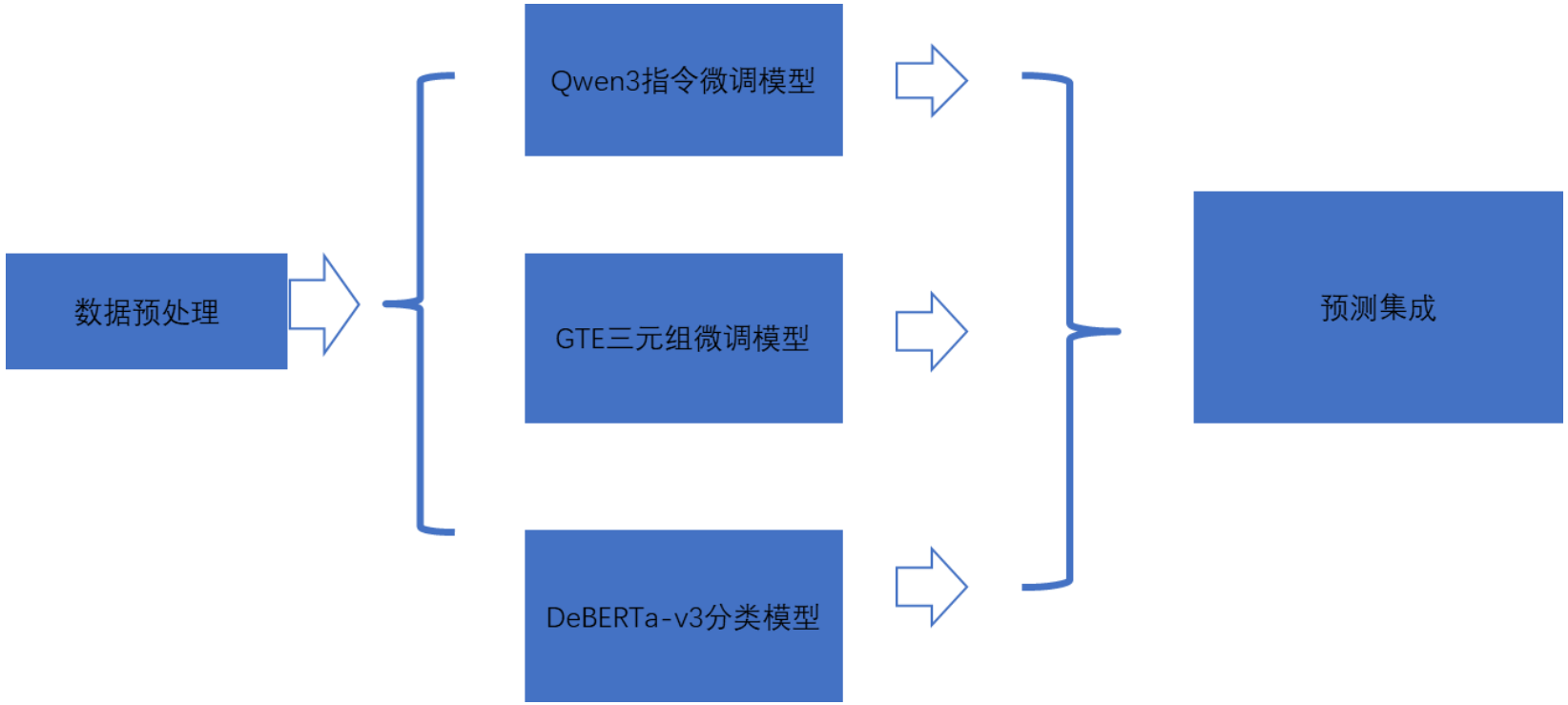


https://developers.google.com/machine-learning/crash-course/classification/roc-and-auc?hl=zh-cn

算法思路

方案概览

本方案采用**三模型集成**的，旨在精准、鲁棒地检测 Reddit 评论是否违反社区规则。该方案融合了大语言模型（LLM） 的深层上下文理解能力、语义相似度模型的度量学习能力以及序列分类模型的强判别能力。最终，通过对三个模型输出进行加权排名融合生成最终的违规概率，以实现最佳的检测效果。



算法思路

测试集使用：内含标签

	row_id	body	rule	subreddit	positive_example_1	positive_example_2	negative_example_1	negative_example_2
0	2029	NEW RAP GROUP 17. CHECK US OUT https://soundcl...	No Advertising: Spam, referral links, unsolici...	hiphopheads	Hey, guys, just wanted to drop in and invite y...	Cum Swallowing Hottie Katrina Kaif Cartoon Xvi...	SD Stream Eng - [Chelsea TV USA](http://soccer...	HD Streams: |[ENG HD Stoke vs Manchester Unite...
1	2030	Make your life comfortable. Get up to 15% Disc...	No legal advice: Do not offer or request legal...	AskReddit	Get a lawyer and get the security camera foota...	That isn't drastic. You tried reaching out to ...	So what are you going to do with the insurance...	It's just for Austria & Germany. If you still ...
2	2031	Kickin' ass and selling underwear!\nJust made ...	No Advertising: Spam, referral links, unsolici...	gonewild	Good story my friend. Check out my blog at ht...	If you know what exactly you need then you don...	CENTIPEDES\n\nSOME BASED PATRIOTS HAVE CREATED...	[So great! Thanks for sharing.](http://www.che...
3	2032	watch hooters best therein http://clickan...	No Advertising: Spam, referral links, unsolici...	personalfinance	Earn 50,000 bonus points with Chase Sapphire P...	Cool, front page! I made this print along with...	[Full HD Movie Online Free](http://www.flickma...	* Karambit Black Pearl\n* 0.02137822 Float (un...
4	2033	bitches for free at this point show all h...	No Advertising: Spam, referral links, unsolici...	Showerthoughts	code free tyrande --->>> [Imgur](http://i.imgu...	My trade link\nhttps://steamcommunity.com/trad...	**HD** [mio Stadium 102 HD](http://www.genti....	Infographics is an incredible method for showi...
5	2034	Top 10 Pentakills in August https://youtu.be/Z...	No Advertising: Spam, referral links, unsolici...	leagueoflegends	code free tyrande --->>> [Imgur](http://i.imgu...	see tits sweet at this point http://sh.or...	its my asreddit https://www.youtube.com/watch?...	click here for more videos[Russian mafia messe...
6	2035	WIN A SAMSUNG SMARTWATCH!\n\nhttp://www.innomd...	No Advertising: Spam, referral links, unsolici...	AskReddit	[ENJOY 50,000 BONUS POINTS after $4,000 spent ...	We have high quality Canadian Polymer banknote...	i was on the same ios and try this jailbreak m...	We're streaming Pokemon Veitnamese Crystal RIG...
7	2036	[This mixtape is lit FAM!](https://beholdthear...	No Advertising: Spam, referral links, unsolici...	BlackPeopleTwitter	watch good one hooters there http://https://x5...	bad pussies waiting for you herein http://gour...	[Full HD Movie Online Free](http://www.flickma...	* Computers :http://livematchstreamz.blogspot....
8	2037	must be watch movie https://sites.google.com/s...	No Advertising: Spam, referral links, unsolici...	movies	see there hooters sweet http://url.t.aylo...	Sign up with my link and get $1,000 of free pr...	The fact that there are pro and cons to everyt...	Oh that's interesting. You should check out my...
9	2038	Free paypal cards here!! https://www.pointspri...	No Advertising: Spam, referral links, unsolici...	pics	she will come your home open her legs with an...	Old and young your girl-next-door make sex wi...	i think the conjuring 2 is not bad movie ,The...	wow, she's so beautyful\n>[Sakura Vietnam](htt...

- 将train.csv和test.csv中提供的positive_example（标签=1）和negative_example（标签=0）也作为训练样本，极大地扩充了训练集。
- 在构建训练样本时，从两个正/负示例中随机选择一个，增加数据多样性。

```
for violation_type in ["positive", "negative"]:
    for i in range(1, 3):
        sub_dataset = test_dataset[["rule", "subreddit", "positive_example_1", "positive_example_2", "negative_example_1", "negative_example_2"]
        body_col = f"{violation_type}_example_{i}"
        other_positive_col = f"{violation_type}_example_{3-i}"
        sub_dataset["body"] = sub_dataset[body_col]
        sub_dataset[f"{violation_type}_example"] = sub_dataset[other_positive_col]
        anti_violation_type = "negative" if violation_type == "positive" else "positive"
        sub_dataset[f"{anti_violation_type}_example"] = np.where(np.random.rand(len(sub_dataset)) < 0.5, sub_dataset[f"{anti_violation_type}_exa
        sub_dataset["rule_violation"] = 1 if violation_type == "positive" else 0
        sub_dataset.drop(columns=["positive_example_1", "positive_example_2", "negative_example_1", "negative_example_2"], inplace=True)
        merge.append(sub_dataset)
return pd.concat(merge, axis=0).drop_duplicates(ignore_index=True)
```

```
for _, row in tqdm(test_df.iterrows(), total=len(test_df), desc="Processing test rows"):
    rule = cleaner(str(row['rule']))

    pos_examples = []  # Will contain compliant comments (rule-aligned)
    neg_examples = []  # Will contain violating comments (rule-misaligned)

    for neg_col in ['negative_example_1', 'negative_example_2']:  # Compliant → triplet positive
        if pd.notna(row[neg_col]):
            pos_examples.append(cleaner(str(row[neg_col])))

    for pos_col in ['positive_example_1', 'positive_example_2']:  # Violating → triplet negative
        if pd.notna(row[pos_col]):
            neg_examples.append(cleaner(str(row[pos_col])))
```

算法思路

模型1：Qwen-4B（上下文理解与生成式判决）

模型选型：Qwen3-4B-base，一个40亿参数的大语言模型，具备出色的指令遵循和上下文学习能力。

核心方法：指令微调（Instruction Fine-tuning）+ QLoRA

提示工程（Prompt Engineering）：构建包含丰富上下文的Few-shot提示，引导模型进行“in-context”学习。提示结构如下：

{系统指令}
Subreddit: r/{社区板块}
Rule: {规则文本}
Examples:
1) {违规示例} -> Violation: Yes
2) {合规示例} -> Violation: No
Comment: {待测评论} -> Violation:

```
def build_prompt(row):
    return f"""{system_prompt}
Subreddit: r/{row["subreddit"]}
Rule: {row["rule"]}
Examples:
1) {row["positive_example"]}
{judge_words} Yes
2) {row["negative_example"]}
{judge_words} No
Comment: {row["body"]}
{judge_words}"""
```

训练技术：

QLoRA：采用4-bit NF4量化和LoRA（秩r=32），在极大降低显存占用的同时高效微调模型。

DeepSpeed ZeRO-2：通过accelerate工具实现分布式训练，优化内存和计算效率。

推理优化：

vLLM引擎：实现极高吞吐量的推理，支持LoRA动态加载。

约束解码：使用MultipleChoiceLogitsProcessor强制模型输出为"Yes"或"No"，确保结果的有效性。

多GPU并行：利用multiprocessing将推理任务分配至多个GPU，进一步加速。

算法思路

模型2：GTE（基于对比学习的语义度量）

模型选型：GTE-base-en-v1.5，一个强大的句子嵌入（Sentence Embedding）模型。

核心方法：三元组损失微调（Triplet Loss Fine-tuning）

三元组构建：此为本模型的关键。我们构建三元组(anchor, positive, negative)来学习规则与评论间的语义关系：

Anchor：规则文本
Positive（正样本）：合规的评论（即原始数据中的negative_example）
Negative（负样本）：违规的评论（即原始数据中的positive_example）
通过这种方式，模型学习将规则的嵌入向量“拉近”合规评论，“推远”违规评论。

推理逻辑：

1. 为每条规则，计算其所有“违规示例”的平均嵌入向量，作为“违规中心点”。
2. 同理，计算所有“合规示例”的平均嵌入向量，作为“合规中心点”。
3. 对于待测评论，计算其嵌入向量与这两个中心点的欧氏距离。
4. 违规分数 = dist(评论, 合规中心) - dist(评论, 违规中心)。评论离违规中心越近、离合规中心越远，分数越高。

算法思路

模型3：DeBERTa-v3（基于深度语言模型的强判别器）

模型选型：DeBERTa-v3-base，在NLU任务上表现卓越的判别式模型。

核心方法：序列分类 + 多种高级训练技巧

输入构建：将规则和评论拼接，并附加上下文信息：{规则文本} [SEP] {评论文本} {URL语义}。其中，url_to_semantics函数提取URL中的关键信息（如域名、路径），作为额外特征。

模型架构创新：

- 注意力池化头（Attention Head）：替代传统的[CLS]池化，对模型最后两层的隐藏状态进行加权平均，动态捕捉最重要的token信息。
- 多Dropout集成（MCDropout）：在分类头设置5个不同概率的Dropout层，推理时取其平均输出，等效于轻量级的模型集成，提升鲁棒性。

高级训练策略：

- 对抗训练（FGM）：在训练过程中对词嵌入层添加微小扰动，迫使模型学习更具泛化性的特征。
- 指数移动平均（EMA）：平滑训练过程，获得更优的最终模型。在评估和预测时使用，通常能获得更好的性能和稳定性。
- 标签平滑（Label Smoothing）:防止模型对标签过于自信，减轻过拟合。

算法思路

加权排名融合（Weighted Rank Blending）

因为我们的目标是最大化AUC，而AUC本身就是一个基于排序的指标，所以我们选择了一种直接操作排序信息（Rank）的融合方法。加权排名融合能有效消除不同模型间的尺度和分布差异，确保每个模型的“排序能力”都能被公平、有效地利用，从而获得比直接分数融合更稳定、更优越的最终AUC表现。

1. **获取排名**：对每个模型（DeBERTa, GTE, LLM）的预测结果按row_id排序，并计算其分数的排名（Rank）。
2. **归一化排名**：将排名转换为[0, 1]区间的比例值（rank / (N+1)）。
3. **加权平均**：对归一化后的排名进行加权求和，得到最终的融合分数。

    权重分配：LLM(0.7), DeBERTa(0.5), GTE(0.3)

    blend = 0.7 * rank_llm + 0.5 * rank_deberta + 0.3 * rank_gte

权重考量：给予LLM最高权重，因其强大的上下文和Few-shot学习能力；DeBERTa作为强判别器次之；GTE作为度量学习模型提供稳定但相对基础的信号，权重最低。

算法思路

算法亮点总结

- **多范式融合**：结合了生成式（Qwen）、对比学习（GTE）和判别式（DeBERTa）三大AI范式，实现了优势互补。
- **资源高效**：通过QLoRA、DeepSpeed、vLLM和梯度检查点等技术，使得训练和推理大参数模型成为可能，兼顾了性能与效率。
- **数据充分利用**：将数据集中的“示例”转化为“训练样本”，是一种高效的伪标签和数据增强策略。
- **鲁棒性设计**：在DeBERTa模型中集成了对抗训练、EMA、多Dropout等多种前沿技术，显著提升了模型的泛化能力和稳定性。
- **智能融合策略**：采用加权排名融合，有效结合了各模型的优势，避免了单一模型可能存在的偏见，是提升最终性能的关键。